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To cite this article: Chengxi Lu *et al* 2026 *J. Phys.: Conf. Ser.* **3229** 012029

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# An Improved YOLOv8-Based Approach for Insulator Fault Detection

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**Abstract.** Accurate identification of insulator faults is a critical guarantee for the secure operation of modern power transmission systems, especially amid the escalating demand for automated and real-time inspection. Deep learning-based detection methods have markedly boosted inspection efficiency, yet striking a practical balance between detection precision and computational efficiency remains a tough challenge in intricate outdoor environments. To address this issue, this study proposes an enhanced insulator fault detection framework built upon YOLOv8. A lightweight MobileNetV3 backbone is adopted to reduce computational burden while preserving essential feature extraction capability. In addition, a novel XXC2f module is introduced to jointly exploit spatial-domain structures and frequency-domain characteristics, thereby improving robustness to background clutter and scale variation. Furthermore, the SIOU loss is employed to guide bounding box regression with improved stability for slender and irregular targets. Experimental results demonstrate that the proposed approach achieves superior detection accuracy with reduced model complexity, making it suitable for real-time insulator inspection applications.

## 1. Introduction

As power systems continue to expand in scale and complexity, higher requirements have been imposed on the efficiency and real-time capability of power equipment inspection. In transmission lines, insulators play a crucial role by providing mechanical support for conductors and ensuring electrical isolation from the ground[1, 2]. The operational state of insulators is therefore closely related to the reliability and safety of power grids. Nevertheless, conventional insulator inspection methods still rely predominantly on manual visual observation, which is characterized by low efficiency and high maintenance costs. In addition, manual inspection is easily influenced by environmental conditions and human factors in complex operating scenarios, often resulting in missed detections and false alarms. Consequently, traditional inspection approaches can no longer satisfy the demands of automated and intelligent inspection under smart grid development.

In recent years, deep learning-driven object detection technologies have shown promise in insulator detection and fault recognition, with existing methods generally falling into two categories, two-stage and one-stage models.

Two-stage detection models first produce region proposals and then conduct target classification and bounding box refinement on them. While ensuring precise localization, this



approach demands significant computational resources. Typical models in this category include R-CNN[3], Fast R-CNN[4] and Faster R-CNN[5]. Some studies have applied these models to insulator detection by optimizing anchor settings or backbone network structures[6, 7]. However, the multi-stage processing pipeline limits the inference speed, making these models unsuitable for real-time inspection scenarios.

By contrast, single-stage object detection models streamline the processes of object localization and classification into a unified regression framework, thereby achieving superior computational efficiency and demonstrating stronger suitability for real-time detection applications. Notable representatives of this model category include the YOLO family[8] and SSD[9]. Recent advancements have focused on incorporating multi-scale feature fusion techniques and attention mechanisms into lightweight YOLO architectures, significantly bolstering the model's detection robustness when operating in environments with complex backgrounds[10, 11, 12, 13]. However, current single-stage models still struggle to strike a balance between detection accuracy and real-time performance, especially when dealing with multi-scale insulator defects.

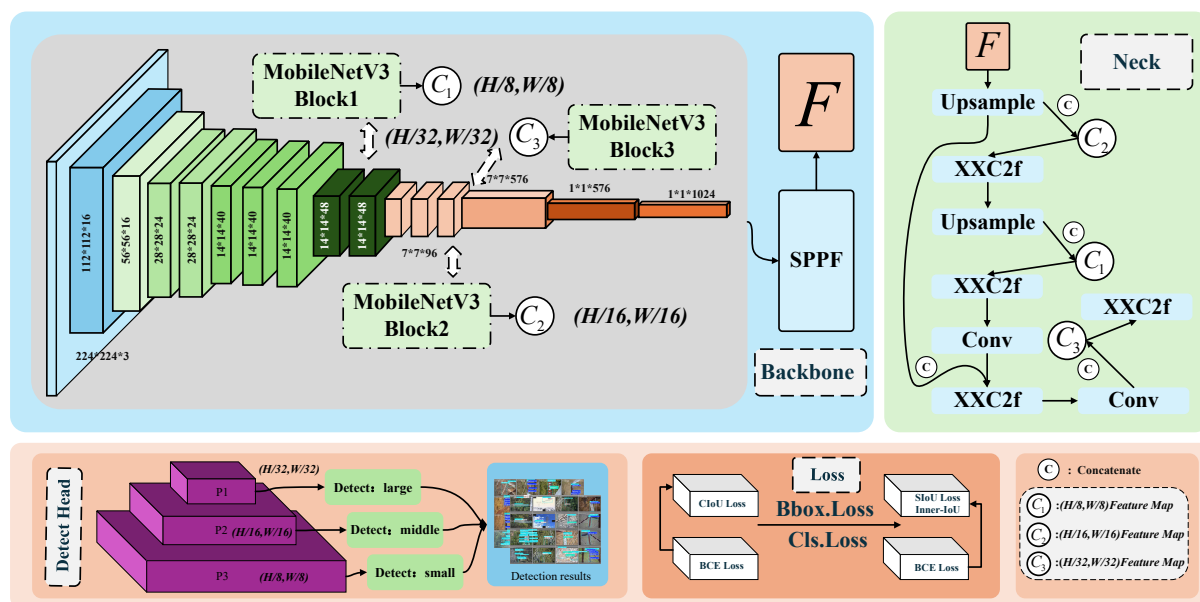
Despite progress, accurately identifying insulator types and detecting faults is still tough due to intricate backgrounds and the need to balance accuracy with real-time performance, which limits further detection improvements. Therefore, developing a dedicated and efficient detection algorithm for insulator fault detection is of substantial research importance. To this end, this paper adopts the YOLOv8 model as the baseline and proposes a series of targeted improvements, which can be summarized as follows:

- A lightweight MobileNetV3[14] component is integrated into the backbone network to substitute for a portion of the original convolutional layers. This modification effectively reduces computational overhead while preserving the model's capacity for feature extraction and enhancing its inference speed in real-world detection tasks.
- A novel XXC2f module that combines spatial-domain and frequency-domain feature fusion is designed to enhance insulator feature extraction and representation, thereby improving adaptability to multi-scale targets.
- The SIoU loss function is employed in place of the original CIoU loss, leading to improved bounding box regression accuracy and target localization performance.

## 2. Materials and methods

### 2.1. Overall Architecture of the Improved Model

Figure 1 illustrates the overall architecture of the proposed improved YOLOv8-based model. By introducing a more lightweight MobileNetV3 network to replace the original backbone, the computational overhead of the model is effectively reduced. Meanwhile, an improved XXC2f module is designed and incorporated into the neck to replace the original C2f structure. By jointly extracting spatial-domain and frequency-domain features, the proposed XXC2f module effectively compensates for and enhances insulator feature representations under complex scenarios. In addition, to address the instability of the CIoU loss when optimizing objects with extreme aspect ratios and its inability to consider bounding box angle alignment, the SIoU loss function with angle cost constraints is adopted. This provides a more optimal convergence path, thereby further improving bounding box regression accuracy and target localization performance. Experimental results demonstrate that the proposed method significantly enhances detection performance while reducing model complexity, making it well suited for insulator type recognition and fault detection tasks.



**Figure 1.** The overall architecture of the improved YOLOv8 model

### 2.2. MobileNetV3

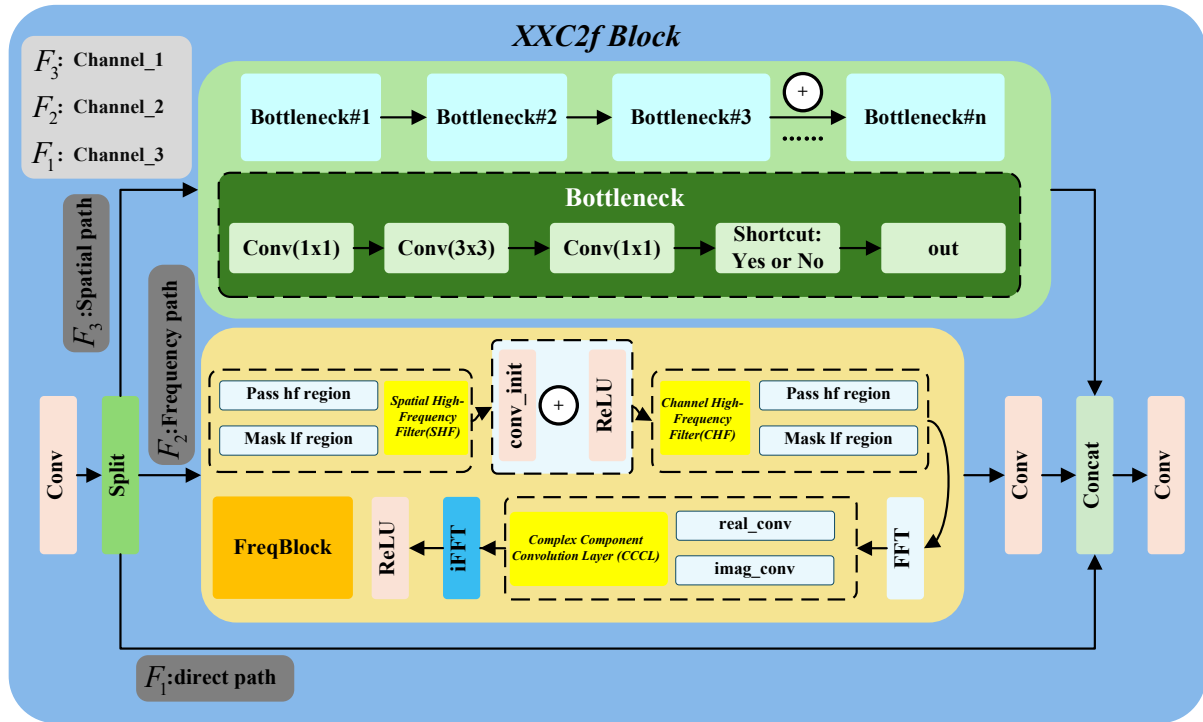
To improve computational efficiency while maintaining effective feature extraction, this study selects MobileNetV3 as the lightweight backbone component.

Unlike conventional bulky convolutional architectures, MobileNetV3 adopts a series of design principles tailored for computational efficiency. These encompass the utilization of depthwise separable convolutions[15], inverted residual blocks featuring linear bottleneck configurations[16], and the integration of squeeze-and-excitation (SE) attention mechanisms[17]. In addition, the hard-swish activation function is introduced to enhance the nonlinear representation ability of the model with minimal computational cost. In the proposed YOLOv8-based framework, part of the original CSPDarknet backbone is replaced with MobileNetV3, leading to a substantial reduction in model parameters and floating-point operations. This modification effectively decreases inference latency and computational complexity, while retaining sufficient feature representation capability for insulator targets. As a result, the lightweight backbone design is particularly suitable for real-time insulator fault detection in practical power inspection scenarios.

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### 2.3. $XXC2f$ block

To enhance feature representation capability for insulator fault detection under complex backgrounds, this paper proposes a lightweight multi-branch feature fusion module termed **XXC2f**, which integrates spatial-domain and frequency-domain information within the YOLOv8 framework. Compared with the original C2f structure, **XXC2f** introduces frequency-domain high-frequency enhancement to improve sensitivity to fine-grained features such as cracks, breakages, and edge defects of insulators.



**Figure 2.** Block diagram of XXC2f

**2.3.1. Spatial and Channel High-frequency Enhancement** In transmission line inspection scenarios, discriminative information of insulator faults is mainly concentrated in high-frequency components. For an input feature map  $I \in \mathbb{R}^{C \times H \times W}$ , where  $C$ ,  $H$ , and  $W$  represent channels, height, and width, spatial-domain high-frequency enhancement is first performed by applying a Fourier transform along the spatial dimensions:

$$I_s^h = \text{Re}(\text{iFFT}_s(\psi_s(\text{FFT}_s(I)))) + I \quad (1)$$

where  $\text{FFT}_s(\cdot)$  and  $\text{iFFT}_s(\cdot)$  represent the Fast Fourier Transform and its inverse across spatial dimensions, respectively, and  $\text{Re}(\cdot)$  extracts the real part.  $\psi_s(\cdot)$  represents a spatial high-pass filter used to suppress low-frequency components.

Subsequently, channel-domain high-frequency enhancement is conducted as:

$$I_c^h = \text{Re}(\text{iFFT}_c(\psi_c(\text{FFT}_c(I_s^h)))) + I_s^h \quad (2)$$

where  $\text{FFT}_c(\cdot)$  and  $\text{iFFT}_c(\cdot)$  denote Fourier transform operations along the channel dimension, and  $\psi_c(\cdot)$  is a channel-domain high-pass filter. These residual enhancement operations effectively amplify high-frequency components while preserving low-frequency contextual information.

**2.3.2. Frequency-domain Complex Convolution** To further exploit frequency-domain characteristics, the enhanced feature  $I_c^h$  is transformed into the frequency domain using a 2D Fourier transform:

$$X = \text{FFT}_s(I_c^h) = X_r + jX_i \quad (3)$$

where  $X_r$  and  $X_i$  denote the real and imaginary components corresponding to amplitude and phase information, respectively. Separate convolution operations are applied to these components:

$$\begin{aligned} Y_r &= W_r * X_r \\ Y_i &= W_i * X_i \end{aligned} \quad (4)$$

where  $W_r$  and  $W_i$  are learnable convolution kernels. The processed components are recombined and transformed back to the spatial domain:

$$I^{out} = \text{Re}(\text{iFFT}_s(Y_r + jY_i)) \quad (5)$$

This mechanism enables adaptive modulation of both amplitude and phase information, enhancing the representation of insulator edge structures and defect contours.

**2.3.3. Multi-branch Feature Fusion Structure** In terms of the structural design, the XXC2f module uses a three-branch parallel framework. For the input feature map  $F$ , a  $1 \times 1$  convolution operation is executed first, and the output feature map is equally divided into three sub-feature maps:

$$[F_1, F_2, F_3] = \text{Split}(\text{Conv}_{1 \times 1}(F)) \quad (6)$$

Here,  $F_1$  is directly propagated to preserve original semantic information,  $F_2$  is fed into the frequency enhancement branch, and  $F_3$  is processed by a series of spatial-domain bottleneck blocks. The outputs of the three branches are concatenated and fused:

$$F_{cat} = \text{Concat}(F_1, F_2, F_3) \quad (7)$$

followed by a  $1 \times 1$  convolution to obtain the final output:

$$F_{out} = \text{Conv}_{1 \times 1}(F_{cat}) \quad (8)$$

By jointly modeling spatial and frequency-domain features, the proposed XXC2f module enhances robustness and discrimination for insulator fault detection while maintaining a lightweight computational structure.

#### 2.4. SIOU Loss

To improve localization stability for insulator faults characterized by small sizes and elongated shapes, this work employs the SIOU loss[18] for bounding box regression. Unlike the CIoU loss, which primarily focuses on the overlap ratio, central distance, and aspect ratio alignment during optimization, the SIOU loss explicitly addresses orientation-related misalignment issues. This approach enables a more efficient regression trajectory for bounding box refinement. Given a predicted bounding box  $B$  and the corresponding ground-truth box  $B^{gt}$ , the SIOU loss is formulated as

$$\mathcal{L}_{\text{SIOU}} = 1 - \text{IoU} + \frac{\Delta + \Omega}{2} \quad (9)$$

where  $\Delta$  and  $\Omega$  represent the center distance cost and shape cost, respectively. By incorporating an angle-related constraint into the regression process, SIOU reduces unnecessary optimization directions and accelerates convergence, which is particularly beneficial for insulator fault detection under complex spatial layouts and extreme aspect-ratio variations.

### 3. Experiments

#### 3.1. Experimental environment configuration

In the training phase, the input image resolution is set to  $640 \times 640$ , and the model is trained for 100 epochs with a batch size of 64. The initial learning rate is set to 0.01 with a decay factor of 0.1, and the SGD optimizer is used for parameter updating. The number of data-loading workers is set to 8. Mosaic data augmentation is enabled in the early training stage and disabled in the last 10 epochs to improve training stability. In addition, an early stopping strategy was applied, where training was terminated if no performance improvement was observed on the validation set for 50 consecutive epochs.

#### 3.2. Dataset

The dataset developed in this research is compiled from manual inspection logs and on-site UAV-captured images of transmission lines, encompassing a diverse range of shooting perspectives, varying lighting conditions, and intricate background settings. This dataset effectively mirrors the real-world inspection scenarios encountered in engineering applications for transmission lines. The dataset includes four insulator condition categories—normal, flashover, missing, and damaged—representing typical fault types in power transmission systems. A total of 3,000 images were collected and augmented using data augmentation techniques, and the dataset was divided into training, validation, and test sets with a ratio of 7:2:1 for model training and performance evaluation.

#### 3.3. Experimental results analysis

To evaluate the effectiveness of the proposed method, comparative experiments were conducted between the improved model (OURS) and several mainstream lightweight object detection networks under identical experimental settings on the same insulator fault dataset. The quantitative results are presented in Table 1, and qualitative detection results are shown in Fig. 3.

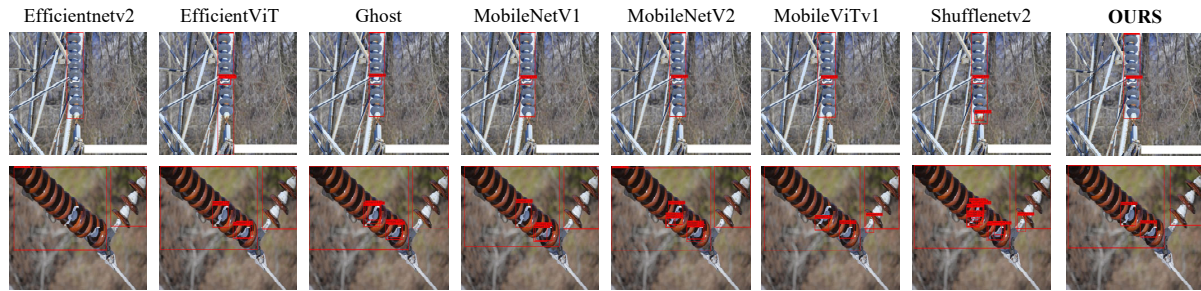
**Table 1.** Comparison of Different Object Detection Network Models

| Models         | Precision    | Recall       | mAP@50       | mAP@50:95    | GFLOPs     | FPS( $f \cdot s^{-1}$ ) |
|----------------|--------------|--------------|--------------|--------------|------------|-------------------------|
| YOLOv8n        | 0.891        | 0.805        | 0.859        | 0.525        | 8.1        | 55.80                   |
| Efficientnetv2 | 0.474        | 0.390        | 0.370        | 0.244        | 2.7        | 60.51                   |
| EfficientViT   | 0.851        | 0.701        | 0.750        | 0.455        | 9.8        | 16.39                   |
| Ghost          | 0.852        | 0.712        | 0.771        | 0.473        | 5.0        | 80.01                   |
| MobileNetV1    | 0.862        | 0.735        | 0.792        | 0.490        | 15.7       | 80.38                   |
| MobileNetV2    | 0.857        | 0.741        | 0.796        | 0.495        | 10.1       | 67.06                   |
| MobileViTv1    | 0.842        | 0.717        | 0.763        | 0.481        | 9.9        | 60.25                   |
| Shufflenetv2   | 0.862        | 0.695        | 0.764        | 0.461        | 7.4        | 37.48                   |
| <b>OURS</b>    | <b>0.921</b> | <b>0.825</b> | <b>0.880</b> | <b>0.578</b> | <b>7.2</b> | <b>82.96</b>            |

As shown in Table 1, the proposed method achieves the highest precision (0.921) and recall (0.825), which proves its excellent ability to reduce false positives and missed detections. In addition, the OURS model achieves mAP@50 and mAP@50:95 values of 0.880 and 0.578 respectively, indicating the best overall detection performance among all compared methods. Although certain comparative models exhibit reduced computational complexity, they suffer from a significant decline in detection accuracy. Consequently, these models are unsuitable for practical power transmission line inspection scenarios that demand high reliability.

In terms of efficiency, the proposed model maintains a low computational cost of 7.2 GFLOPs while achieving an inference speed of 82.96 FPS, outperforming the baseline YOLOv8n and

most lightweight models. The visualization results further demonstrate that the proposed method provides more stable and accurate detection under complex backgrounds and multi-scale insulator scenarios.



**Figure 3.** Validation results of different algorithms

### 3.4. Ablation Study

To further analyze the contribution of each proposed component, ablation experiments were conducted by progressively introducing the MobileNetV3 backbone, the XXC2f module, and the SIOU loss function. The results are presented in Table 2.

**Table 2.** The results of ablation study

| MobileNetV3 | XXC2f | SIOU | Precision    | Recall       | mAP@50       | mAP@50:95    | GFLOPs     | FPS( $f \cdot s^{-1}$ ) |
|-------------|-------|------|--------------|--------------|--------------|--------------|------------|-------------------------|
| ✓           |       |      | 0.881        | 0.804        | 0.856        | 0.526        | 7.2        | 78.60                   |
|             | ✓     |      | 0.860        | 0.783        | 0.835        | 0.519        | 8.8        | 65.86                   |
|             |       | ✓    | 0.900        | 0.815        | 0.864        | 0.521        | 8.1        | 60.80                   |
| ✓           | ✓     |      | 0.910        | 0.794        | 0.859        | 0.525        | 7.2        | 79.82                   |
| ✓           | ✓     | ✓    | <b>0.921</b> | <b>0.825</b> | <b>0.880</b> | <b>0.578</b> | <b>7.2</b> | <b>82.96</b>            |

When the model solely incorporates the MobileNetV3 lightweight backbone, there is a substantial reduction in the overall network's computational complexity (to 7.2 GFLOPs). Meanwhile, the model retains a competitive detection accuracy, thereby validating the efficacy of the lightweight feature extraction approach proposed in this research. Introducing the XXC2f module alone enhances feature representation capability but slightly increases computational cost. Similarly, replacing CIOU with SIOU improves localization performance, leading to higher Precision and mAP values, though with a moderate increase in inference cost.

When MobileNetV3 and XXC2f are jointly applied, the model achieves a better balance between accuracy and efficiency. Finally, by integrating all three improvements—MobileNetV3, XXC2f, and SIOU—the proposed model achieves the best overall performance, with Precision, Recall, mAP@50, and mAP@50:95 reaching 0.921, 0.825, 0.880, and 0.578, respectively, while maintaining low computational complexity (7.2 GFLOPs) and high inference speed (82.96 FPS).

The experimental findings presented in this study confirm that every enhanced module introduced in this paper contributes positively to boosting the model's overall performance. Furthermore, the collaborative implementation of these modules not only substantially elevates detection accuracy but also maintains the model's real-time detection capabilities.

## 4. Conclusion

This paper proposes an improved YOLOv8-based insulator fault detection method for power inspection scenarios. By introducing a lightweight MobileNetV3 backbone, a spatial-frequency

fusion XXC2f module and the SIOU loss function, the proposed model effectively improves detection accuracy while reducing computational complexity. Experiments show it outperforms mainstream lightweight object detectors in key metrics like precision, recall, and mAP, while maintaining low GFLOPs and high real-time inference speed. This approach effectively balances accuracy and efficiency, making it well-suited for real-time insulator fault detection in practical power inspection scenarios.

## Acknowledgments

This work was supported by Sichuan Science and Technology Program (No.2026YFHZ0262).

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